

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Eighth Semester B.E. (Mechanical) (Credit System) Degree Examinations**

May - June 2023

19MEE65 - NON DESTRUCTIVE TESTING

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.*

		Marks	BT*	CO*	PO*
Unit – I					
1.	a) Explain visual inspection and various equipment's used to aid visual inspection.	10	L*2	1	1
	b) Explain the principle of magnetic particle inspection and state its advantages.	10	L2	1	1
2.	a) Explain with sketch the eddy current inspection system.	10	L2	2	1
	b) Explain the various types of inspection coils used in eddy current inspection with neat sketches.	10	L2	2	1
3.	a) Explain different inspection methods used in thermal inspection.	10	L2	2	1
	b) Describe the basic principle of computed tomography.	10	L2	2	1
Unit – II					
4.	a) Discuss in brief the basic Equipment's and operation variables for ultrasonic inspection.	12	L2	3	1
	b) Explain different types of ultrasonic waves.	8	L2	3	1
5.	a) Explain with sketch recording films used in radiographic inspection.	10	L2	4	1
	b) Explain real time radiography and film radiography.	10	L2	4	1
6.	a) Discuss in brief the following with respect to radiographic inspection. i) Production of X-rays ii) Gama rays iii) Properties of X-Rays and Gamma Rays.	12	L2	4	1
	b) Discuss in brief application of radiographic inspection in industry.	8	L2	4	1
Unit – III					
7.	a) Explain the principle involved in acoustic emission inspection. State its applications.	10	L2	5	1
	b) Describe briefly acoustic emission sensor with suitable sketch.	10	L2	5	1
8.	a) Compare the characteristics of acoustic emission inspection with other inspection methods.	5	L2	5	1
	b) Explain the generic block diagram of a four-channel acoustic emission system.	10	L2	5	1
	c) Discuss in brief the propagation of acoustic emission waves.	5	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
